

while Iteration Statement

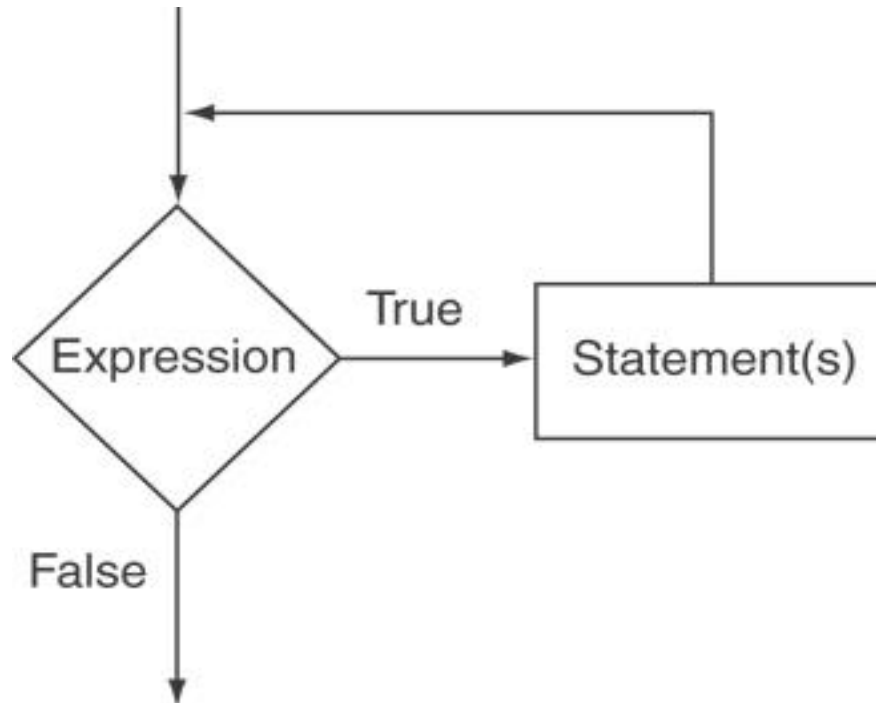
ELEC1006: ENGINEERING COMPUTING

Iteration Statement (loops)

- Loops are used to repeat a statement a certain number of times when a condition is met.
- There are three types of loops:
 - *while* loop. ←
 - *do-while* loop.
 - *for* loop.

The *while* Loop

- The format is:
while (expression or condition) { Statement(s) }
– repeat while “expression or condition” is **true**.
- The logical flow of while loop:

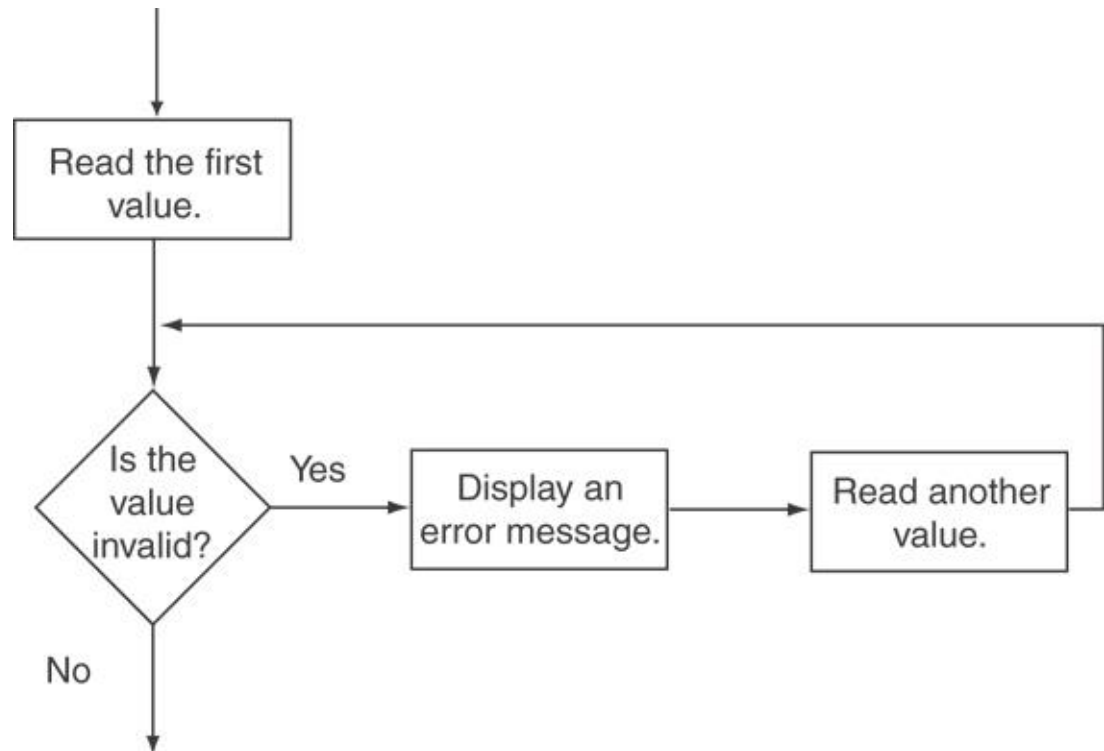


Example

```
// custom countdown using while
#include <iostream>
using namespace std;
int main ()
{
    int n;
    cout << "Enter the starting number: ";
    cin >> n;
    while (n>0) // Don't put ; at the end
    {
        cout << n << ", ";
        --n; // Decrement n by 1 in every loop
    }
    cout << "FIRE!\n";
    return 0;
}
```

Input validation

```
cout << "Enter a number less than 10: ";  
cin >> number;  
while (number >= 10)  
{  
    cout << "Invalid Entry!"  
        << "Enter a number less than 10: ";  
    cin >> number;  
}
```



Tips: When to use *while*

- Use *while* statement when the number of times the loop is required to run is unknown.
- Use *while* (1) to form an infinite loop especially when using a simple program that runs for long periods of time, e.g. mobile phone call:

while (1)

```
{  
    read_input();      // 1) Get network data  
    process_data();    // 2) Convert into sound signal  
    output_data();     // 3) Play signal on speaker  
    if call_end()  
        break;        // Break the while loop if call ends  
}
```

More info

- [1] cplusplus.com: Statements and flow control
<https://cplusplus.com/doc/tutorial/control/>
- [2] learncpp.com: 5.5 – While statements
<https://www.learncpp.com/cpp-tutorial/55-while-statements/>